

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250279873

Kind Code

A1

Publication Date

September 04, 2025

Inventor(s)

Takahashi; Yuki et al.

TERMINAL, BASE STATION AND COMMUNICATION METHOD

Abstract

A terminal includes: a communication unit configured to perform an uplink or downlink communication by using a frequency band in which aggregation is performed by carrier aggregation; and a control unit configured to expect, in a case where scheduling is performed by using a unit including a plurality of component carriers, that a time domain resource allocation is to be configured for each component carrier among the plurality of component carriers or for each unit including a plurality of component carriers in the aggregated frequency band.

Inventors:	Takahashi; Yuki (Chiyoda-ku, Tokyo, JP), Kumagai; Shinya (Chiyoda-ku, Tokyo, JP), Yoshioka; Shohei (Chiyoda-ku, Tokyo, JP), Okamura; Masaya (Chiyoda-ku, Tokyo, JP), Okano; Mayuko (Chiyoda-ku, Tokyo, JP), Nagata; Satoshi (Chiyoda-ku, Tokyo, JP)
Applicant:	NTT DOCOMO, INC. (Tokyo, JP)
Family ID:	88518198
Assignee:	NTT DOCOMO, INC. (Tokyo, JP)
Appl. No.:	18/859327
Filed (or PCT Filed):	April 28, 2022
PCT No.:	PCT/JP2022/019453

Publication Classification

Int. Cl.: H04L5/00 (20060101); H04W72/0446 (20230101)

U.S. Cl.:

CPC H04L5/0098 (20130101); H04W72/0446 (20130101);

Background/Summary

FIELD OF THE INVENTION

[0001] The present invention relates to a terminal, a base station and a communication method in a wireless communication system.

BACKGROUND OF THE INVENTION

[0002] Regarding NR (New Radio) (also referred to as “5G”), which is a successor system to LTE (Long Term Evolution), technologies have been discussed which satisfy the following requirements: a high capacity system, high data transmission rate, low delay, simultaneous connection of multiple terminals, low cost, power saving, etc. (for example, Non-Patent Document 1).

[0003] In addition, in NR, as a continued discussion from LTE, a carrier aggregation (CA) function using a wide band in order to allocate data resources is being discussed. With the carrier aggregation function, wide band data resources can be allocated by bundling a plurality of component carriers (CC).

CITATION LIST

Non-Patent Document

[0004] Non-Patent Document 1: 3GPP TS 38.300 V16.8.0 (2021-12)

SUMMARY OF THE INVENTION

Technical Problem

[0005] In future systems (for example, NR release 18 and 6G that is a successor system of NR), there is a possibility that a more flexible and efficient resource allocation will be required.

However, with the conventional carrier aggregation function, there has been a problem that the resource allocation overhead is large due to the necessity of scheduling data resources for each of the bundled plurality of component carriers.

[0006] The present invention has been made in view of the foregoing and is intended to reduce the overhead for the resource allocation.

Solution to Problem

[0007] According to the disclosed technique, a terminal is provided. The terminal includes: a communication unit configured to perform an uplink or downlink communication by using a frequency band in which aggregation is performed by carrier aggregation; and a control unit configured to expect, in a case where scheduling is performed by using a unit including a plurality of component carriers, that a time domain resource allocation is to be configured for each component carrier among the plurality of component carriers or for each unit including a plurality of component carriers in the aggregated frequency band.

Advantageous Effects of Invention

[0008] According to the disclosed technique, a technology is provided that enables reduction of the resource allocation overhead.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. 1 is a drawing for describing a wireless communication system related to an embodiment of the present invention.

[0010] FIG. 2 is a first drawing illustrating an example of a structure of a virtual CC related to an embodiment of the present invention.

[0011] FIG. 3 is a second drawing illustrating an example of a structure of a virtual CC related to an embodiment of the present invention.

[0012] FIG. 4 is a drawing illustrating an example of the conventional TDRA table.

[0013] FIG. 5 is a first drawing for describing the conventional SLIV.

[0014] FIG. 6 is a second drawing for describing the conventional SLIV.

[0015] FIG. 7 is a drawing illustrating an example of a functional structure of a base station related to an embodiment of the present invention.

[0016] FIG. 8 is a drawing illustrating an example of a functional structure of a terminal related to an embodiment of the present invention,

[0017] FIG. 9 is a drawing illustrating an example of a hardware structure of the base station or the terminal related to an embodiment of the present invention.

[0018] FIG. 10 is a drawing illustrating an example of a structure of a vehicle related to an embodiment of the present invention.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

[0019] In the following, referring to the drawings, one or more embodiments of the present invention will be described. It should be noted that the embodiments described below are examples. Embodiments of the present invention are not limited to the following embodiments.

[0020] In operations of a wireless communication system according to an embodiment of the present invention, a conventional technique will be used when it is appropriate. With respect to the above, for example, the conventional techniques are related to, but not limited to, the existing LTE. Further, it is assumed that the term “LTE” used in the present specification has, unless otherwise specifically mentioned, a broad meaning including a scheme of LTE Advanced and a scheme after LTE-Advanced (e.g., NR).

[0021] Furthermore, in one or more embodiments described below, terms that are used in the existing LTE are used, such as SS (Synchronization signal), PSS (Primary SS), SSS (Secondary SS), PBCH (Physical broadcast channel), PRACH (Physical random access channel), PDCCH (Physical Downlink Control Channel), PDSCH (Physical Downlink Shared Channel), PUCCH (Physical Uplink Control Channel), PUSCH (Physical Uplink Shared Channel), etc. The above described terms are used for the sake of description convenience. Signals, functions, etc., which are similar to the above-described terms, may be referred to as different names. Further, terms that are used in NR and correspond to the above-described terms are NR-SS, NR-PSS, NR-SSS, NR-PBCH, NR-PRACH, etc. However, even when a signal is used for NR, there may be cases in which the signal is not referred to as “NR-”.

[0022] In addition, in an embodiment of the present invention, the duplex method may be a TDD (Time Division Duplex) method, an FDD (Frequency Division Duplex) method, or any other method (e.g., Flexible Duplex, or the like).

[0023] Further, in an embodiment of the present invention, the expression, radio (wireless) parameters are “configured (set)” may mean that a predetermined value is pre-configured, or may mean that a radio parameter indicated by a base station 10 or a terminal 20 is configured. (System Configuration)

[0024] FIG. 1 is a drawing illustrating a wireless communication system related to an embodiment of the present invention.

[0025] As illustrated in FIG. 1, the wireless communication system according to an embodiment of the present invention includes a base station 10 and a terminal 20. In FIG. 1, a single base station 10 and a single terminal 20 are illustrated as an example. There may be a plurality of base stations 10 and a plurality of terminals 20.

[0026] The base station 10 is a communication device that provides one or more cells and performs wireless communication with the terminal 20. Physical resources of radio signals may be defined in the time domain and the frequency domain, the time domain may be defined by the number of OFDM (Orthogonal Frequency Division Multiplexing) symbols, and the frequency domain may be defined by the number of sub-carriers or resource blocks. Further, a TTI (Transmission Time Interval) in the time domain may be a slot, or the TTI may be a subframe.

[0027] The base station **10** transmits a synchronization signal and system information to the terminal **20**. The synchronization signal is, for example, an NR-PSS and an NR-SSS. The system information is transmitted via, for example, an NR-PBCH, and may be referred to as broadcast information. The synchronization signal and the system information may be referred to as an SSB (SS/PBCH block). As shown in FIG. **1**, the base station **10** transmits a control signal or data in DL (Downlink) to the terminal **20** and receives a control signal or data in UL (Uplink) from the terminal **20**. The base station **10** and terminal **20** are capable of transmitting and receiving a signal by performing the beamforming. Further, the base station **10** and the terminal **20** can both apply MIMO (Multiple Input Multiple Output) communication to DL or UL. Further, the base station **10** and the terminal **20** may both perform communications via a secondary cell (SCell: Secondary Cell) and a primary cell (PCell: Primary Cell) using CA (Carrier Aggregation). In addition, the terminal **20** may perform communications via a primary cell of the base station **10** and a primary secondary cell group cell (PSCell: Primary SCG Cell) of another base station **10** using DC (Dual Connectivity).

[0028] The terminal **20** may be a communication apparatus that includes a wireless communication function such as a smartphone, a mobile phone, a tablet, a wearable terminal, a communication module for M2M (Machine-to-Machine), or the like. As shown in FIG. **1**, the terminal **20** uses various communication services provided by the wireless communication system by receiving control signals or data in DL from the base station **10** and transmitting control signals or data in UL to the base station **10**. In addition, the terminal **20** receives various reference signals transmitted from the base station **10** and performs measurement of the propagation path quality based on the reception result of the reference signals. Note that the terminal **20** may be referred to as a UE, and the base station **10** may be referred to as a gNB.

[0029] In addition, in NR, as a continued discussion from LTE, a carrier aggregation function using a wide band in order to allocate data resources is being discussed. With the carrier aggregation function, wide band data resources can be allocated by bundling a plurality of component carriers. (Problem of Conventional Technique)

[0030] With the conventional carrier aggregation function, there has been a problem that the resource allocation overhead is large due to the necessity of scheduling data resources for each of the bundled plurality of component carriers.

[0031] Accordingly, in order to solve the above-described conventional problem, a method can be considered in which the terminal allocates resources by using a scheduling unit with a granularity that is different from that of the component carriers.

[0032] A framework for performing scheduling or aggregation with a granularity that is different from that of the component carriers is defined as a frequency fragmentation.

[0033] In addition, in the carrier aggregation, an operation of performing aggregation with a granularity that is different from that of the component carriers is defined as a non-continuous carrier aggregation.

[0034] In addition, in the carrier aggregation (non-continuous carrier aggregation), an operation of scheduling with a granularity that is different from that of the component carriers is defined as a non continuous scheduling.

[0035] The above described granularity different from that of the component carriers may be a virtual CC unit, a BWP (Bandwidth Part) unit, a PRB (Physical Resource Block), or a PRB set unit,

[0036] Here, the virtual CC means a set of carriers in which all or a part of frequency resources included in each component carrier from among a plurality of component carriers are bundled.

[0037] For example, the virtual CC may be expected to include a plurality of BWPs,

[0038] FIG. **2** is a first drawing illustrating an example of a structure of a virtual CC related to an embodiment of the present invention. The virtual CC #*i* illustrated in FIG. **2** is a set of carriers in which BWP #*a* and BWP #*b* included in respective component carriers from among a plurality of component carriers (CC #**0** and CC #**1**) are bundled.

[0039] In addition, the virtual CC may be expected to include a plurality of PRBs or a PRB set, [0040] FIG. 3 is a second drawing illustrating an example of a structure of a virtual CC related to an embodiment of the present invention. The virtual CC #i illustrated in FIG. 3 is a set of carriers in which a plurality of PRBs included in respective component carriers from among a plurality of component carriers (CC #0 and CC #1) are bundled. It is to be noted that the plurality of PRBs or PRB set may be included in one or more BWPs.

[0041] Hereinafter, a CC before bundling is referred to as an actual CC, and a CC after bundling is referred to as a virtual CC or a nominal CC. It is to be noted that the names are not limited to the above.

(Conventional Time Domain Resource Allocation)

[0042] Next, the conventional time domain resource allocation (TDRA) will be described.

Allocation candidates are configured as a table (for example, “pdsch-TimeDomainAllocationList”) via RRC and the terminal determines a time resource to be allocated from among the allocation candidates.

[0043] FIG. 4 is a drawing illustrating an example of the conventional TDRA table. Up to 16 candidates/entries are configured in a TDRA table. Each table includes a mapping type, K.sub.0 in a case of PDSCH and K.sub.2 in a case of PUSCH, and SLIV (Start and Length Indicator Value).

[0044] In a case where RRC is not provided, one of the four default tables is applied. In addition, one of the entries in the TDRA table is selected by using the TDRA field via DCI. The bit length of TDRA is calculated according to $\log_2 I$. Here, I is the number of entries, For example, in a case of 16 entries, the bit length of TDRA is $\log_2 16 = 4$ bits.

[0045] FIG. 5 is a first drawing for describing the conventional SLIV. SLIV includes an allocation of the starting index and the length index. “S” indicates the starting index of the allocated PDSCH. “L” indicates the length of the allocated PDSCH.

[0046] FIG. 6 is a second drawing for describing the conventional SLIV. As illustrated in FIG. 6, the values that “S” and “L” can take are defined in accordance with the PDSCH mapping type.

[0047] SLIV is calculated as follows.

TABLE-US-00001 [Expression 1] if $(L - 1) \leq 7$ then SLIV = $14 \cdot \text{Math.} (L - 1) + S$ else
SLIV = $14 \cdot \text{Math.} (14 - L + 1) + (14 - 1 - S)$ where $0 < L \leq 14 - S$

[0048] Conventionally, there is a problem that the time domain resource allocation in the frequency fragmentation is unclear. As a result, there is a problem that there is a possibility that the time domain resource allocation in the frequency fragmentation cannot be properly performed.

Overview of Embodiment

[0049] Accordingly, in an embodiment of the present invention, the method of time domain resource allocation in the frequency fragmentation will be described.

<Option 1>

[0050] The terminal 20 may expect that TDRA is to be separately performed for each of the BWPs included in a virtual CC. The terminal 20 may perform the time resource allocation for each of the BWPs that are configured for actual CCs included in a virtual CC.

[0051] The terminal 20 may expect that the TDRA table (for example, “pdsch-TimeDomainAllocationList”) is to be configured for each of the two or more BWPs configured for actual CCs included in a virtual CC. In addition, the configuration values in each of the TDRA tables may be different or may be the same.

[0052] The TDRA table may include one or more of one of or a plurality of information items

below. [0053] Starting time and time length (for example, SLIV) [0054] Interval from PDCCH to the starting time of PDSCH/PUSCH (for example, K.sub.0 or K.sub.1) [0055] PDSCH/PUSCH mapping type

[0056] In addition, the terminal 20 may expect that the starting time of PDSCH/PUSCH is to be determined as follows.

[00001]
$$n \cdot \frac{2^{\mu\text{PDSCH}}}{2^{\mu\text{PDCCH}}} + K_0$$
 [Expression2]

[0057] Here, n is a slot of the scheduling DCI. K.sub.0 is a value based on the numerology of PDSCH. μPDSCH and μPDCCH are corresponding numerologies indicating the subcarrier spacing configurations of PDSCH and PDCCH.

[00002]
$$n \cdot \frac{2^{\mu\text{PUSCH}}}{2^{\mu\text{PDCCH}}} + K_1$$
 [Expression3]

[0058] Here, n is a slot of the scheduling DCI. K.sub.1 is a value based on the numerology of PDSCH. μPDSCH and μPDCCH are corresponding numerologies indicating the subcarrier spacing configurations of PDSCH and PDCCH.

[0059] This option is useful for, but is not limited to, a case in which a separate BWP is configured for each actual CC.

<Option 2>

[0060] The terminal **20** may expect that TDRA is to be performed for the entire virtual CC. The terminal **20** may expect that the TDRA table (for example, “pdsch-TimeDomainAllocationListForVCC”) is to be configured for the BWP that is configured for the virtual CC.

[0061] The TDRA table may include one or more of one of or a plurality of information items below. [0062] Starting time and time length (for example, SLIV) [0063] Interval from PDCCH to the starting time of PDSCH/PUSCH (for example, K.sub.0 or K.sub.1) [0064] PDSCH/PUSCH mapping type

[0065] In addition, the terminal **20** may expect that the starting time of PDSCH/PUSCH is to be determined as follows.

[00003]
$$n \cdot \frac{2^{\mu\text{PDSCH}}}{2^{\mu\text{PDCCH}}} + K_0$$
 [Expression4]

[0066] Here, n is a slot of the scheduling DCI. K.sub.0 is a value based on the numerology of PDSCH. μPDSCH and μPDCCH are corresponding numerologies indicating the subcarrier spacing configurations of PDSCH and PDCCH.

[00004]
$$n \cdot \frac{2^{\mu\text{PUSCH}}}{2^{\mu\text{PDCCH}}} + K_1$$
 [Expression5]

[0067] Here, n is a slot of the scheduling DCI. K.sub.1 is a value based on the numerology of PDSCH. μPDSCH and μPDCCH are corresponding numerologies indicating the subcarrier spacing configurations of PDSCH and PDCCH.

[0068] In addition, in a case where SCSs of the actual CCs are different from each other, the terminal **20** may expect that μPDSCH , μPDCCH , and μPUSCH are to be determined based on a reference SCS. The terminal **20** may expect that the reference SCS is the minimum value, the maximum value, or the average value of the SCSs of the actual CCs in which PDSCH/PDCCH/PUSCH are to be scheduled, may expect that the default value is defined in the technical specifications, or may expect that the reference SCS is configured by an upper layer parameter.

[0069] This option is useful for, but is not limited to, a case where one BWP is configured for the entire virtual CC.

[0070] According to an embodiment of the present invention, the time domain resource allocation can be properly performed in the frequency fragmentation.

(Device Configuration)

[0071] Next, a functional configuration example of the base station **10** and the terminal **20** for performing the processes and operations described above will be described.

<Base Station **10**>

[0072] FIG. 7 is a diagram illustrating an example of a functional configuration of the base station **10**. As shown in FIG. 7, the base station **10** includes a transmission unit **110**, a reception unit **120**, a configuration unit **130**, and a control unit **140**. The functional configuration illustrated in FIG. 7 is

merely an example. Functional divisions and names of functional units may be anything as long as operations according to an embodiment of the present invention can be performed. Further, the transmission unit **110** and the reception unit **120** may be combined and may be referred to as a communication unit.

[0073] The transmission unit **110** includes a function for generating a signal to be transmitted to the terminal **20** side and transmitting the signal wirelessly. The reception unit **120** includes a function for receiving various signals transmitted from the terminal **20** and acquiring, for example, information of a higher layer from the received signals. Further, the transmission unit **110** has a function to transmit NR-PSS, NR-SSS, NR-PBCH, DL/UL control signals, DCI via PDCCH, data via PDSCH, and the like, to the terminal **20**.

[0074] The configuration unit **130** stores preset configuration information and various configuration information items to be transmitted to the terminal **20** in a storage device included in the configuration unit **130** and reads the preset configuration information from the storage apparatus if necessary.

[0075] The control unit **140** performs scheduling of the terminal **20** for DL reception or UL transmission, via the transmission unit **110**. In addition, the control unit **140** includes a function of performing LBT. The functional units related to signal transmission in the control unit **140** may be included in the transmission unit **110**, and the functional units related to signal reception in the control unit **140** may be included in the reception unit **120**. Further, the transmission unit **110** may be referred to as a transmitter, and the reception unit **120** may be referred to as a receiver.

<Terminal **20**>

[0076] FIG. **8** is a diagram illustrating an example of a functional configuration of the terminal **20**. As shown in FIG. **8**, the terminal **20** includes a transmission unit **210**, a reception unit **220**, a configuration unit **230**, and a control unit **240**. The functional configuration illustrated in FIG. **8** is merely an example. Functional divisions and names of functional units may be anything as long as operations according to an embodiment of the present invention can be performed. The transmission unit **210** and the reception unit **220** may be combined and may be referred to as a communication unit,

[0077] The transmission unit **210** generates a transmission signal from transmission data and transmits the transmission signal wirelessly. The reception unit **220** receives various signals wirelessly and obtains upper layer signals from the received physical layer signals. In addition, the reception unit **220** has a function of receiving NR-PSS, NR-SSS, NR-PBCH, DL/UL/SL control signals, DCI via PDCCH, data via PDSCH, etc., transmitted from the base station **10**. In addition, for example, with respect to the D2D communications, the transmission unit **210** may transmit, to another terminal **20**, PSCCH (Physical Sidelink Control Channel), PSSCH (Physical Sidelink Shared Channel), PSDCH (Physical Sidelink Discovery Channel), PSBCH (Physical Sidelink Broadcast Channel), etc., and the reception unit **120** may receive, from the other terminal **20**, PSCCH, PSSCH, PSDCH, or PSBCH.

[0078] The configuration unit **230** stores various configuration information items received from the base station **10** or the other terminal by the reception unit **220** in the storage device included in the configuration unit **230**, and reads them from the storage device as necessary. In addition, the configuration unit **230** also stores pre-configured configuration information. The control unit **240** controls the terminal **20**. In addition, the control unit **240** includes a function of performing LBT.

[0079] A terminal according to an embodiment of the present invention may be configured as a terminal described in each item below. In addition, a communication method below may be performed.

<Configuration Related to an Embodiment of the Present Invention>

(First Item)

[0080] A terminal including: [0081] a communication unit configured to perform an uplink or downlink communication by using a frequency band in which aggregation is performed by carrier

aggregation; and [0082] a control unit configured to expect, in a case where scheduling is performed by using a unit including a plurality of component carriers, that a time domain resource allocation is to be configured for each component carrier among the plurality of component carriers or for each unit including a plurality of component carriers in the aggregated frequency band.

(Second Item)

[0083] The terminal as described in the first item, wherein [0084] the control unit expects that the time domain resource allocation is to be separately configured for each bandwidth part configured for a corresponding component carrier included in the unit, the unit including the plurality of component carriers.

(Third Item)

[0085] The terminal as described in the first item, wherein [0086] the control unit expects that the time domain resource allocation is to be configured for a bandwidth part that is configured for the unit including the plurality of component carriers.

(Fourth Item)

[0087] A base station comprising: [0088] a communication unit configured to perform a communication with a terminal by using a frequency band in which aggregation is performed by carrier aggregation; and [0089] a control unit configured to configure, in a case where scheduling is performed by using a unit including a plurality of component carriers, a time domain resource allocation for each component carrier among the plurality of component carriers or for each unit including a plurality of component carriers in the aggregated frequency band.

(Fifth Item)

[0090] A communication method performed by a terminal, the communication method including: [0091] performing an uplink or downlink communication by using a frequency band in which aggregation is performed by carrier aggregation; and [0092] expecting, in a case where scheduling is performed by using a unit including a plurality of component carriers, that a time domain resource allocation is to be configured for each component carrier among the plurality of component carriers or for each unit including a plurality of component carriers in the aggregated frequency band.

[0093] According to any one of the above-described configurations, a technique that enables reduction of resource allocation overhead can be provided. According to the first item, in the aggregated frequency band, in a case where scheduling is performed by using a unit including a plurality of component carriers, the time domain resource allocation can be expected to be configured for each of a plurality of component carriers or for each unit including a plurality of component carriers. According to the second item, the time domain resource allocation can be expected to be separately configured for each bandwidth part configured for a corresponding component carrier included in the unit, the unit including the plurality of component carriers. According to the third item, the time domain resource allocation can be expected to be configured for a bandwidth part that is configured for the unit including the plurality of component carriers.

(Hardware Structure)

[0094] In the above block diagrams used for describing an embodiment of the present invention (FIG. 7 and FIG. 8), functional unit blocks are shown. The functional blocks (function units) are realized by a freely-selected combination of hardware and/or software. Further, realizing means of each functional block is not limited in particular. In other words, each functional block may be realized by a single apparatus in which multiple elements are coupled physically and/or logically, or may be realized by two or more apparatuses that are physically and/or logically separated and are physically and/or logically connected (e.g., wired and/or wireless). The functional blocks may be realized by combining the above-described one or more apparatuses with software.

[0095] Functions include, but are not limited to, judging, determining, calculating, processing, deriving, investigating, searching, checking, receiving, transmitting, outputting, accessing, resolving, selecting, establishing, comparing, assuming, expecting, and deeming; broadcasting,

notifying, communicating, forwarding, configuring, reconfiguring, allocating, mapping, and assigning, etc. For example, a functional block (component) that functions to transmit is called a transmitting unit or a transmitter. In either case, as described above, the implementation method is not particularly limited.

[0096] For example, the base station **10**, the terminal **20**, etc., according to an embodiment of the present disclosure may function as a computer for processing the radio communication method of the present disclosure. FIG. **9** is a drawing illustrating an example of hardware structures of the base station **10** and the terminal **20** according to an embodiment of the present invention. Each of the above-described base station **10** and terminal **20** may physically be a computer device including a processor **1001**, a storage device **1002**, an auxiliary storage device **1003**, a communication device **1004**, an input device **1005**, an output device **1006**, a bus **1007**, etc.

[0097] It should be noted that, in the descriptions below, the term “device” can be read as a circuit, a device, a unit, etc. The hardware structures of the base station **10** and the terminal **20** may include one or more of each of the devices illustrated in the figure, or may omit some devices.

[0098] Each function in the base station **10** and the terminal **20** is realized by having the processor **1001** perform an operation by reading predetermined software (programs) onto hardware such as the processor **1001** and the storage device **1002**, and by controlling communication by the communication device **1004** and controlling at least one of reading and writing of data in the storage device **1002** and the auxiliary storage device **1003**.

[0099] The processor **1001** controls the entire computer by, for example, controlling the operating system. The processor **1001** may include a central processing unit (CPU) including an interface with a peripheral apparatus, a control apparatus, a calculation apparatus, a register, etc. For example, the above-described control unit **140**, control unit **240**, and the like, may be implemented by the processor **1001**.

[0100] Further, the processor **1001** reads out onto the storage device **1002** a program (program code), a software module, or data from the auxiliary storage device **1003** and/or the communication device **1004**, and performs various processes according to the program, the software module, or the data. As the program, a program is used that causes the computer to perform at least a part of operations according to an embodiment of the present invention described above. For example, the control unit **140** of the base station **10** illustrated in FIG. **7** may be realized by control programs that are stored in the storage device **1002** and are executed by the processor **1001**. Further, for example, the control unit **240** of the terminal **20** illustrated in FIG. **8** may be realized by control programs that are stored in the storage device **1002** and are executed by the processor **1001**. The various processes have been described to be performed by a single processor **1001**. However, the processes may be performed by two or more processors **1001** simultaneously or sequentially. The processor **1001** may be implemented by one or more chips. It should be noted that the program may be transmitted from a network via a telecommunication line.

[0101] The storage device **1002** is a computer-readable recording medium, and may include at least one of a ROM (Read Only Memory), an EPROM (Erasable Programmable ROM), an EEPROM (Electrically Erasable Programmable ROM), a RAM (Random Access Memory), etc. The storage device **1002** may be referred to as a register, a cache, a main memory, etc. The storage device **1002** is capable of storing programs (program codes), software modules, or the like, that are executable for performing communication processes according to an embodiment of the present invention.

[0102] The auxiliary storage device **1003** is a computer-readable recording medium, and may include at least one of, for example, an optical disk such as a CD-ROM (Compact Disc ROM), a hard disk drive, a flexible disk, a magneto optical disk (e.g., compact disc, digital versatile disc, Blu-ray (registered trademark) disk), a smart card, a flash memory (e.g., card, stick, key drive), a floppy (registered trademark) disk, a magnetic strip, etc. The above recording medium may be a database including the storage device **1002** and/or the auxiliary storage device **1003**, a server, or any other appropriate medium.

[0103] The communication device **1004** is hardware (transmission and reception device) for communicating with computers via at least one of a wired network or a wireless network, and may be referred to as a network device, a network controller, a network card, a communication module, etc. The communication device **1004** may comprise a high frequency switch, duplexer, filter, frequency synthesizer, or the like, for example, to implement at least one of a frequency division duplex (FDD) or a time division duplex (TDD). For example, the transmitting/receiving antenna, the amplifier unit, the transmitting/receiving unit, the transmission line interface, and the like, may be implemented by the communication device **1004**. The transmitting/receiving unit may be physically or logically divided into a transmitting unit and a receiving unit.

[0104] The input device **1005** is an input device that receives an external input (e.g., keyboard, mouse, microphone, switch, button, sensor). The output device **1006** is an output device that outputs something to the outside (e.g., display, speaker, LED lamp). It should be noted that the input device **1005** and the output device **1006** may be integrated into a single device (e.g., touch panel).

[0105] Further, the apparatuses including the processor **1001**, the storage device **1002**, etc. are connected to each other via the bus **1007** used for communicating information. The bus **1007** may include a single bus, or may include different buses between the apparatuses.

[0106] Further, each of the base station **10** and terminal **20** may include hardware such as a microprocessor, a digital signal processor (DSP), an ASIC (Application Specific Integrated Circuit), a PLD (Programmable Logic Device), a FPGA (Field Programmable Gate Array), etc., and a part or all of each functional block may be realized by the hardware. For example, the processor **1001** may be implemented by at least one of the above hardware elements.

[0107] FIG. **10** shows an example of a configuration of a vehicle **2001**. As shown in FIG. **10**, the vehicle **2001** includes a drive unit **2002**, a steering unit **2003**, an accelerator pedal **2004**, a brake pedal **2005**, a shift lever **2006**, a front wheel **2007**, a rear wheel **2008**, an axle **2009**, an electronic control unit **2010**, various sensors **2021-2029**, an information service unit **2012**, and a communication module **2013**. The aspects/embodiments described in the present disclosure may be applied to a communication device mounted in the vehicle **2001**, and may be applied to, for example, the communication module **2013**.

[0108] The drive unit **2002** may include, for example, an engine, a motor, and a hybrid of an engine and a motor. The steering unit **2003** includes at least a steering wheel and is configured to steer at least one of the front wheel and the rear wheel, based on the operation of the steering wheel operated by the user.

[0109] The electronic control unit **2010** includes a microprocessor **2031**, a memory (ROM, RAM) **2032**, and a communication port (IO port) **2033**. The electronic control unit **2010** receives signals from the various sensors **2021-2029** provided in the vehicle **2001**. The electronic control unit **2010** may be referred to as an ECU (Electronic control unit).

[0110] The signals from the various sensors **2021** to **2029** include a current signal from a current sensor **2021** which senses the current of the motor, a front or rear wheel rotation signal acquired by a revolution sensor **2022**, a front or rear wheel pneumatic signal acquired by a pneumatic sensor **2023**, a vehicle speed signal acquired by a vehicle speed sensor **2024**, an acceleration signal acquired by an acceleration sensor **2025**, an accelerator pedal depression amount signal acquired by an accelerator pedal sensor **2029**, a brake pedal depression amount signal acquired by a brake pedal sensor **2026**, an operation signal of a shift lever acquired by a shift lever sensor **2027**, and a detection signal, acquired by an object detection sensor **2028**, for detecting an obstacle, a vehicle, a pedestrian, and the like.

[0111] The information service unit **2012** includes various devices for providing (outputting) various kinds of information such as driving information, traffic information, and entertainment information, including a car navigation system, an audio system, a speaker, a television, and a radio, and one or more ECUs controlling these devices. The information service unit **2012** provides

various types of multimedia information and multimedia services to the occupants of the vehicle **2001** by using information obtained from the external device through the communication module **2013** or the like.

[0112] The information service unit **2012** may include an input device (e.g., a keyboard, mouse, microphone, switch, button, sensor, touch panel, etc.) that accepts an external input, or may include an output device (e.g., a display, speaker, LED lamp, touch panel, etc.) that implements an external output.

[0113] A driving support system unit **2030** includes: various devices for providing functions of preventing accidents and reducing a driver's operating burden such as a millimeter wave radar, a LIDAR (Light Detection and Ranging), a camera, a positioning locator (e.g., GNSS, etc.), map information (e.g., high definition (HD) map, autonomous vehicle (AV) map, etc.), a gyro system (e.g., IMU (Inertial Measurement Unit), INS (Inertial Navigation System), etc.), an AI (Artificial Intelligence) chip, an AI processor; and one or more ECUs controlling these devices. In addition, the driving support system unit **2030** transmits and receives various types of information via the communication module **2013** to realize a driving support function or an autonomous driving function.

[0114] The communication module **2013** may communicate with the microprocessor **2031** and components of the vehicle **2001** via a communication port. For example, the communication module **2013** transmits and receives data via a communication port **2033**, to and from the drive unit **2002**, the steering unit **2003**, the accelerator pedal **2004**, the brake pedal **2005**, the shift lever **2006**, the front wheel **2007**, the rear wheel **2008**, the axle **2009**, the microprocessor **2031** and the memory (ROM, RAM) **2032** in the electronic control unit **2010**, and sensors **2021** to **2029** provided in the vehicle **2001**.

[0115] The communication module **2013** is a communication device that can be controlled by the microprocessor **2031** of the electronic control unit **2010** and that is capable of communicating with external devices. For example, various kinds of information are transmitted to and received from external devices through radio communication. The communication module **2013** may be internal to or external to the electronic control unit **2010**. The external devices may include, for example, a base station, a mobile station, or the like.

[0116] The communication module **2013** may transmit, to an external device by using wireless communications, at least one of: a signal from the above-described various sensors **2021** to **2029** that is input to the electronic control unit **2010**; information that is obtained based on the signal; or information based on an input obtained from outside (user) via the information service unit **2012**. The electronic control unit **2010**, the various sensors **2021** to **2029**, the information service unit **2012**, or the like, may be referred to as an input unit for receiving an input. For example, PUSCH transmitted by the communication module **2013** may include information based on the above-described input.

[0117] The communication module **2013** receives various types of information (traffic information, signal information, inter-vehicle information, etc.) transmitted from the external devices and displays the received information on the information service unit **2012** provided in the vehicle **2001**. The information service unit **2012** may be referred to as an output unit for outputting information (for example, outputting information to devices such as a display or a speaker, based on PDSCH (or data/information decoded from the PDSCH) received by the communication module **2013**).

[0118] In addition, the communication module **2013** stores the various types of information received from the external devices in the memory **2032** available to the microprocessor **2031**. Based on the information stored in the memory **2032**, the microprocessor **2031** may control the drive unit **2002**, the steering unit **2003**, the accelerator pedal **2004**, the brake pedal **2005**, the shift lever **2006**, the front wheel **2007**, the rear wheel **2008**, the axle **2009**, the sensors **2021-2029**, etc., mounted in the vehicle **2001**.

Supplement of Embodiment

[0119] As described above, one or more embodiments have been described. The present invention is not limited to the above embodiments. A person skilled in the art should understand that there are various modifications, variations, alternatives, replacements, etc., of the embodiments. In order to facilitate understanding of the present invention, specific values have been used in the description. However, unless otherwise specified, those values are merely examples and other appropriate values may be used. The division of the described items may not be essential to the present invention. The things that have been described in two or more items may be used in a combination if necessary, and the thing that has been described in one item may be appropriately applied to another item (as long as there is no contradiction). Boundaries of functional units or processing units in the functional block diagrams do not necessarily correspond to the boundaries of physical parts. Operations of multiple functional units may be physically performed by a single part, or an operation of a single functional unit may be physically performed by multiple parts. The order of sequences and flowcharts described in an embodiment of the present invention may be changed as long as there is no contradiction. For the sake of description convenience, the base station **10** and the terminal **20** have been described by using functional block diagrams. However, the apparatuses may be realized by hardware, software, or a combination of hardware and software. The software executed by a processor included in the base station **10** according to an embodiment of the present invention and the software executed by a processor included in the terminal **20** according to an embodiment of the present invention may be stored in a random access memory (RAM), a flash memory, a read only memory (ROM), an EPROM, an EEPROM, a register, a hard disk (HDD), a removable disk, a CD-ROM, a database, a server, or any other appropriate recording medium.

[0120] Further, information indication may be performed not only by methods described in an aspect/embodiment of the present specification but also a method other than those described in an aspect/embodiment of the present specification. For example, the information indication may be performed by physical layer signaling (e.g., DCI (Downlink Control Information), UCI (Uplink Control Information)), upper layer signaling (e.g., RRC (Radio Resource Control) signaling, MAC (Medium Access Control) signaling, broadcast information (MIB (Master Information Block), SIB (System Information Block))), other signals, or combinations thereof. Further, RRC signaling may be referred to as an RRC message. The RRC signaling may be, for example, an RRC connection setup message, an RRC connection reconfiguration message, or the like.

[0121] Each aspect/embodiment described in the present disclosure may be applied to at least one of a system using LTE (Long Term Evolution), LTE.Math.A (LTE-Advanced), SUPER 3G, IMT-Advanced, 4G (4th generation mobile communication system), 5G (5th generation mobile communication system), 6th generation mobile communication system (6G), xth generation mobile communication system (xG) (xG (x is, for example, an integer or a decimal)), FRA (Future Radio Access), NR (new Radio), New radio access (NX), Future generation radio access (FX), W-CDMA (registered trademark), GSM (registered trademark), CDMA2000, UMB (Ultra Mobile Broadband), IEEE 802.11 (Wi-Fi (registered trademark)), IEEE 802.16 (WiMAX (registered trademark)), IEEE 802.20, UWB (Ultra-WideBand), Bluetooth (registered trademark), and other appropriate systems, and a next generation system enhanced, modified, developed, or defined therefrom. Further, multiple systems may also be applied in combination (e.g., at least one of LTE or LTE-A combined with 5G, etc.).

[0122] The order of processing steps, sequences, flowcharts or the like of an aspect/embodiment described in the present specification may be changed as long as there is no contradiction. For example, in a method described in the present specification, elements of various steps are presented in an exemplary order. The order is not limited to the presented specific order.

[0123] The particular operations, that are supposed to be performed by the base station **10** in the present specification, may be performed by an upper node in some cases. In a network including one or more network nodes including the base station **10**, it is apparent that various operations

performed for communicating with the terminal **20** may be performed by the base station **10** and/or another network node other than the base station **10** (for example, but not limited to, MME or S-GW). According to the above, a case is described in which there is a single network node other than the base station **10**. However, a combination of multiple other network nodes may be considered (e.g., MME and S-GW).

[0124] The information or signals described in this disclosure may be output from a higher layer (or lower layer) to a lower layer (or higher layer). The information or signals may be input or output through multiple network nodes.

[0125] The input or output information may be stored in a specific location (e.g., memory) or managed using management tables. The input or output information may be overwritten, updated, or added. The information that has been output may be deleted. The information that has been input may be transmitted to another apparatus.

[0126] A decision or a determination in an embodiment of the present invention may be realized by a value (0 or 1) represented by one bit, by a boolean value (true or false), or by comparison of numerical values (e.g., comparison with a predetermined value).

[0127] Software should be broadly interpreted to mean, whether referred to as software, firmware, middle ware, microcode, hardware description language, or any other name, instructions, instruction sets, codes, code segments, program codes, programs, subprograms, software modules, applications, software applications, software packages, routines, subroutines, objects, executable files, executable threads, procedures, functions, and the like.

[0128] Further, software, instructions, information, and the like may be transmitted and received via a transmission medium. For example, in the case where software is transmitted from a website, server, or other remote source using at least one of wired line technologies (such as coaxial cable, fiber optic cable, twisted pair, digital subscriber line (DSL), etc.) or wireless technologies (infrared, microwave, etc.), at least one of these wired line technologies or wireless technologies is included within the definition of the transmission medium.

[0129] Information, a signal, or the like, described in the present specification may be represented by using any one of various different technologies. For example, data, an instruction, a command, information, a signal, a bit, a symbol, a chip, or the like, described throughout the present application, may be represented by a voltage, an electric current, electromagnetic waves, magnetic fields, a magnetic particle, optical fields, a photon, or a combination thereof.

[0130] It should be noted that a term used in the present specification and/or a term required for understanding of the present specification may be replaced by a term having the same or similar meaning. For example, a channel and/or a symbol may be a signal (signaling). Further, a signal may be a message. Further, the component carrier (CC) may be referred to as a carrier frequency, cell, frequency carrier, or the like.

[0131] As used in the present disclosure, the terms “system” and “network” are used interchangeably.

[0132] Further, the information, parameters, and the like, described in the present disclosure may be expressed using absolute values, relative values from predetermined values, or they may be expressed using corresponding different information. For example, a radio resource may be what is indicated by an index.

[0133] The names used for the parameters described above are not used as limitations. Further, the mathematical equations using these parameters may differ from those explicitly disclosed in the present disclosure. Because the various channels (e.g., PUCCH, PDCCH) and information elements may be identified by any suitable names, the various names assigned to these various channels and information elements are not used as limitations.

[0134] In the present disclosure, the terms “BS: Base Station”, “Radio Base Station”, “Base Station”, “Fixed Station”, “NodeB”, “eNodeB (eNB)”, “gNodeB (gNB)”, “Access Point”, “Transmission Point”, “Reception Point”, “Transmission/Reception Point”, “Cell”, “Sector”, “Cell

Group”, “Carrier”, “Component Carrier”, and the like, may be used interchangeably. The base station may be referred to as a macro-cell, a small cell, a femtocell, a picocell and the like.

[0135] The base station may accommodate (provide) one or more (e.g., three) cells. In the case where the base station accommodates a plurality of cells, the entire coverage area of the base station may be divided into a plurality of smaller areas, and each smaller area may provide communication services by means of a base station subsystem (e.g., an indoor small base station or a remote Radio Head (RRH)). The term “cell” or “sector” refers to a part or all of the coverage area of at least one of the base station and base station subsystem that provides communication services at the coverage.

[0136] In the present disclosure, a transmission of information from the base station to the terminal may be replaced with an indication of control or operation from the base station to the terminal based on the information.

[0137] In the present disclosure, terms such as “mobile station (MS)”, “user terminal”, “user equipment (UE)”, “terminal”, and the like, may be used interchangeably.

[0138] There is a case in which the mobile station may be referred to, by a person skilled in the art, as a subscriber station, a mobile unit, a subscriber unit, a wireless unit, a remote unit, a mobile device, a wireless device, a wireless communication device, a remote device, a mobile subscriber station, an access terminal, a mobile terminal, a wireless terminal, a remote terminal, a handset, a user agent, a mobile client, a client, or some other appropriate terms.

[0139] At least one of the base station or the mobile station may be referred to as a transmission apparatus, reception apparatus, communication apparatus, or the like. The at least one of the base station or the mobile station may be a device mounted on the mobile station, the mobile station itself, or the like. The mobile station is an object that can move, and the moving speed can be any speed. In addition, a mobile station that is not moving is also included. The mobile station includes, but is not limited to, a vehicle, a transport vehicle, an automobile, a motorcycle, a bicycle, a connected car, an excavator, a bulldozer, a wheel loader, a dump truck, a forklift, a train, a bus, a rear car, a rickshaw, a ship and other watercraft, an airplane, a rocket, an artificial satellite, a drone (registered trademark), a multi copter, a quadcopter, a balloon, and an object mounted thereon. In addition, the mobile station may be a mobile station that autonomously travels based on an operation command. The mobile station may be a vehicle (e.g., a car, an airplane, etc.), an unmanned mobile station (e.g., a drone, an automated vehicle, etc.), or a robot (manned or unmanned). At least one of the base station or the mobile station may include an apparatus that does not necessarily move during communication operations. For example, at least one of the base station or the mobile station may be an IoT (Internet of Things) device such as a sensor.

[0140] Further, the base station in the present disclosure may be read as the user terminal. For example, each aspect/embodiment of the present disclosure may be applied to a configuration in which communications between the base station and the user terminal are replaced by communications between multiple terminals **20** (e.g., may be referred to as D2D (Device-to-Device), V2X (Vehicle-to Everything), etc.). In this case, the function of the base station **10** described above may be provided by the terminal **20**. Further, the phrases “up” and “down” may also be replaced by the phrases corresponding to terminal-to-terminal communication (e.g., “side”). For example, an uplink channel, a downlink channel, or the like, may be read as a sidelink channel.

[0141] Further, the user terminal in the present disclosure may be read as the base station. In this case, the function of the user terminal described above may be provided by the base station.

[0142] The term “determining” used in the present specification may include various actions or operations. The terms “determination” and “decision” may include “determination” and “decision” made with judging, calculating, computing, processing, deriving, investigating, searching (looking up, search, inquiry) (e.g., search in a table, a database, or another data structure), or ascertaining. Further, the “determining” may include “determining” made with receiving (e.g., receiving information), transmitting (e.g., transmitting information), inputting, outputting, or accessing (e.g.,

accessing data in a memory). Further, the “determining” may include a case in which “resolving”, “selecting”, “choosing”, “establishing”, “comparing”, or the like is deemed as “determining”. In other words, the “determining” may include a case in which a certain action or operation is deemed as “determining”. Further, “decision” may be read as “assuming”, “expecting”, or “considering”, etc.

[0143] The term “connected” or “coupled” or any variation thereof means any direct or indirect connection or connection between two or more elements and may include the presence of one or more intermediate elements between the two elements “connected” or “coupled” with each other. The coupling or connection between the elements may be physical, logical, or a combination thereof. For example, “connection” may be read as “access”. As used in the present disclosure, the two elements may be thought of as being “connected” or “coupled” to each other using at least one of the one or more wires, cables, or printed electrical connections and, as a number of non-limiting and non-inclusive examples, electromagnetic energy having wavelengths in the radio frequency region, the microwave region, and the light (both visible and invisible) region.

[0144] The reference signal may be abbreviated as RS or may be referred to as a pilot, depending on the applied standards.

[0145] The description “based on” used in the present specification does not mean “based on only” unless otherwise specifically noted. In other words, the phrase “based on” means both “based on only” and “based on at least”.

[0146] Any reference to an element using terms such as “first” or “second” as used in the present disclosure does not generally limit the amount or the order of those elements. These terms may be used in the present disclosure as a convenient way to distinguish between two or more elements. Therefore, references to the first and second elements do not imply that only two elements may be employed or that the first element must in some way precede the second element.

[0147] “Means” included in the configuration of each of the above apparatuses may be replaced by “parts”, “circuits”, “devices”, etc.

[0148] In the case where the terms “include”, “including” and variations thereof are used in the present disclosure, these terms are intended to be comprehensive in the same way as the term “comprising”. Further, the term “or” used in the present specification is not intended to be an “exclusive or”.

[0149] A radio frame may include one or more frames in the time domain. Each of the one or more frames in the time domain may be referred to as a subframe. The subframe may further include one or more slots in the time domain. The subframe may be a fixed length of time (e.g., 1 ms) independent from the numerology.

[0150] The numerology may be a communication parameter that is applied to at least one of the transmission or reception of a signal or channel. The numerology may indicate at least one of, for example, SubCarrier Spacing (SCS), bandwidth, symbol length, cyclic prefix length, transmission time interval (TTI), number of symbols per TTI, radio frame configuration, specific filtering processing performed by the transceiver in the frequency domain, or specific windowing processing performed by the transceiver in the time domain.

[0151] The slot may include one or more symbols in the time domain, such as OFDM (Orthogonal Frequency Division Multiplexing) symbols, SC-FDMA (Single Carrier Frequency Division Multiple Access) symbols, and the like. The slot may be a time unit based on the numerology.

[0152] The slot may include a plurality of mini slots. Each mini slot may include one or more symbols in the time domain. Further, the mini slot may be referred to as a sub-slot. The mini slot may include fewer symbols than the slot. PDSCH (or PUSCH) transmitted in time units greater than a mini slot may be referred to as PDSCH (or PUSCH) mapping type A. PDSCH (or PUSCH) transmitted using a mini slot may be referred to as PDSCH (or PUSCH) mapping type B.

[0153] A radio frame, a subframe, a slot, a mini slot and a symbol all represent time units for transmitting signals. Different terms may be used for referring to a radio frame, a subframe, a slot,

a mini slot and a symbol, respectively.

[0154] For example, one subframe may be referred to as a transmission time interval (TTI), multiple consecutive subframes may be referred to as a TTI, and one slot or one mini slot may be referred to as a TTI. In other words, at least one of the subframe and the TTI may be a subframe (1 ms) in an existing LTE, a period shorter than 1 ms (e.g., 1-13 symbols), or a period longer than 1 ms. It should be noted that the unit representing the TTI may be referred to as a slot, a mini slot, or the like, rather than a subframe,

[0155] The TTI refers to, for example, the minimum time unit for scheduling in wireless communications. For example, in an LTE system, a base station schedules each terminal **20** to allocate radio resources (such as frequency bandwidth, transmission power, etc. that can be used in each terminal **20**) in TTI units. The definition of TTI is not limited to the above.

[0156] The TTI may be a transmission time unit, such as a channel-encoded data packet (transport block), code block, codeword, or the like, or may be a processing unit, such as scheduling or link adaptation. It should be noted that, when a TTI is provided, the time interval (e.g., the number of symbols) during which the transport block, code block, codeword, or the like, is actually mapped may be shorter than the TTI

[0157] It should be noted that, when one slot or one mini slot is referred to as a TTI, one or more TTIs (i.e., one or more slots or one or more mini slots) may be the minimum time unit for scheduling. Further, the number of slots (the number of mini slots) constituting the minimum time unit of the scheduling may be controlled.

[0158] A TTI having a time length of 1 ms may be referred to as a normal TTI (a TTI in LTE Rel. 8-12), a long TTI, a normal subframe, a long subframe, a slot, and the like. A TTI that is shorter than the normal TTI may be referred to as a shortened TTI, a short TTI, a partial TTI (or fractional TTI), a shortened subframe, a short subframe, a mini slot, a subslot, a slot, or the like.

[0159] It should be noted that the long TTI (e.g., normal TTI, subframe, etc.) may be replaced with a TTI having a time length exceeding 1 ms, and the short TTI (e.g., shortened TTI, etc.) may be replaced with a TTI having a TTI length less than the TTI length of the long TTI and a TTI length greater than 1 ms.

[0160] A resource block (RB) is a time domain and frequency domain resource allocation unit and may include one or more consecutive subcarriers in the frequency domain. The number of subcarriers included in an RB may be the same, regardless of the numerology, and may be 12, for example. The number of subcarriers included in an RB may be determined on the basis of numerology.

[0161] Further, the time domain of an RB may include one or more symbols, which may be 1 slot, 1 mini slot, 1 subframe, or 1 TTI in length. One TTI, one subframe, etc., may each include one or more resource blocks.

[0162] It should be noted that one or more RBs may be referred to as physical resource blocks (PRBs, Physical RBs), sub-carrier groups (SCGs), resource element groups (REGs), PRB pairs, RB pairs, and the like.

[0163] Further, a resource block may include one or more resource elements (RE). For example, 1 RE may be a radio resource area of one sub-carrier and one symbol.

[0164] The bandwidth part (BWP) (which may also be referred to as a partial bandwidth, etc.) may represent a subset of consecutive common RBs (common resource blocks) for a given numerology in a carrier. Here, a common RB may be identified by an index of RB relative to the common reference point of the carrier. A PRB may be defined in a BWP and may be numbered within the BWP.

[0165] BWP may include BWP for UL (UL BWP) and BWP for DL (DL BWP). For a terminal **20**, one or more BWPs may be configured in one carrier.

[0166] At least one of the configured BWPs may be activated, and the terminal **20** may assume that the terminal **20** will not transmit and receive signals/channels outside the activated BWP. It should

be noted that the terms “cell” and “carrier” in this disclosure may be replaced by “BWP.”

[0167] Structures of a radio frame, a subframe, a slot, a mini slot, and a symbol described above are exemplary only. For example, the number of subframes included in a radio frame, the number of slots per subframe or radio frame, the number of mini slots included in a slot, the number of symbols and RBs included in a slot or mini slot, the number of subcarriers included in an RB, the number of symbols in a TTI, the symbol length, the cyclic prefix (CP) length, and the like, may be changed in various ways.

[0168] In the present disclosure, where an article is added by translation, for example “a”, “an”, and “the”, the disclosure may include that the noun following these articles is plural.

[0169] In this disclosure, the term “A and B are different” may mean “A and B are different from each other.” It should be noted that the term “A and B are different” may mean “A and B are different from C.” Terms such as “separated” or “combined” may be interpreted in the same way as the above-described “different”.

[0170] Each aspect/embodiment described in the present specification may be used independently, may be used in combination, or may be used by switching according to operations. Further, notification (transmission/reporting) of predetermined information (e.g., notification (transmission/reporting) of “X”) is not limited to an explicit notification (transmission/reporting), and may be performed by an implicit notification (transmission/reporting) (e.g., by not performing notification (transmission/reporting) of the predetermined information).

[0171] As described above, the present invention has been described in detail. It is apparent to a person skilled in the art that the present invention is not limited to one or more embodiments of the present invention described in the present specification. Modifications, alternatives, replacements, etc., of the present invention may be possible without departing from the subject matter and the scope of the present invention defined by the descriptions of claims. Therefore, the descriptions of the present specification are for illustrative purposes only, and are not intended to be limitations to the present invention,

DESCRIPTION OF THE REFERENCE NUMERALS

[0172] **10** Base station [0173] **110** Transmission unit [0174] **120** Reception unit [0175] **130** Configuration unit [0176] **140** Control unit [0177] **20** Terminal [0178] **210** Transmission unit [0179] **220** Reception unit [0180] **230** Configuration unit [0181] **240** Control unit [0182] **1001** Processor [0183] **1002** Storage device [0184] **1003** Auxiliary storage device [0185] **1004** Communication device [0186] **1005** Input device [0187] **1006** Output device [0188] **2001** Vehicle [0189] **2002** Drive unit [0190] **2003** Steering unit [0191] **2004** Accelerator pedal [0192] **2005** Brake pedal [0193] **2006** Shift lever [0194] **2007** Front wheel [0195] **2008** Rear wheel [0196] **2009** Axle [0197] **2010** Electronic control unit [0198] **2012** Information service unit [0199] **2013** Communication module [0200] **2021** Current sensor [0201] **2022** Revolution sensor [0202] **2023** Pneumatic sensor [0203] **2024** Vehicle speed sensor [0204] **2025** Acceleration sensor [0205] **2026** Brake pedal sensor [0206] **2027** Shift lever sensor [0207] **2028** Object detection sensor [0208] **2029** Accelerator pedal sensor [0209] **2030** Driving support system unit [0210] **2031** Microprocessor [0211] **2032** Memory (ROM, RAM) [0212] **2033** Communication port (IO port)

Claims

1. A terminal comprising: a communication unit configured to perform an uplink or downlink communication by using a frequency band in which aggregation is performed by carrier aggregation; and a control unit configured to expect, in a case where scheduling is performed by using a unit including a plurality of component carriers, that a time domain resource allocation is to be configured for each component carrier among the plurality of component carriers or for each unit including a plurality of component carriers in the aggregated frequency band.
2. The terminal as claimed in claim 1, wherein the control unit expects that the time domain

resource allocation is to be separately configured for each bandwidth part configured for a corresponding component carrier included in the unit including the plurality of component carriers.

3. The terminal as claimed in claim 1, wherein the control unit expects that the time domain resource allocation is to be configured for a bandwidth part that is configured for the unit including the plurality of component carriers.

4. A base station comprising: a communication unit configured to perform a communication with a terminal by using a frequency band in which aggregation is performed by carrier aggregation; and a control unit configured to configure, in a case where scheduling is performed by using a unit including a plurality of component carriers, a time domain resource allocation for each component carrier among the plurality of component carriers or for each unit including a plurality of component carriers in the aggregated frequency band.

5. A communication method performed by a terminal, the communication method comprising: performing an uplink or downlink communication by using a frequency band in which aggregation is performed by carrier aggregation; and expecting, in a case where scheduling is performed by using a unit including a plurality of component carriers, that a time domain resource allocation is to be configured for each component carrier among the plurality of component carriers or for each unit including a plurality of component carriers in the aggregated frequency band.
